



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)

AIOT

15-01-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	D	D	C	B	C	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	C	A	B	C	B	B	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	2	3	30	8	60	250	1	4	4	3

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	A	A	B	C	D	D	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	C	C	D	C	D	B	D	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	7	100	500	7	1	260	10	300	6	3

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	A	C	D	B	A	C	B	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	A	B	B	A	B	C	A	A	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3600	212	6	3	10	7	4	45	6	15

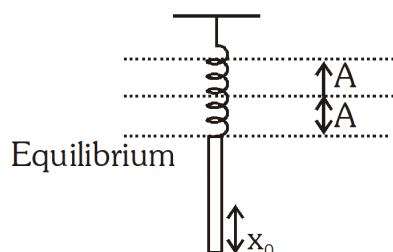
HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (A)

$$KA = \frac{M}{L} x_0 g \quad \dots(i)$$



New equilibrium string will slack if there is compression

$$2A < \frac{Mg}{K} \Rightarrow x_0 < \frac{L}{2}$$

2. Ans (C)

Between 2 nodes all particle vibrate in same phase

3. Ans (A)

For refraction only

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R} \quad \dots(i)$$

$$v = -10 \text{ cm}$$

Differentiate equation (i) w.r.t. time

$$-\frac{\mu_2}{v^2} \frac{dv}{dt} + \frac{\mu_1}{u^2} \frac{du}{dt} = 0$$

$$\frac{dv}{dt} = \frac{\mu_1}{\mu_2} \frac{v^2}{u^2} \frac{du}{dt} = \frac{4}{1 \times 3} \times 1 \times 6 = 8 \text{ m/s}$$

(towards left)

Image after reflection from mirror will create at object it self and after refraction from curved surface it move to rightward with 8 m/s.

$$\text{So } V_{\text{rel}} = 16 \text{ m/s}$$

4. **Ans (D)**

$$\tan A = 4/3; F_A = 40/3 \text{ N} = 13.33 \text{ N}$$

$$\tan B = 2/3; F_B = 20/3 \text{ N} = 6.67 \text{ N}$$

$$\tan C = 1/\sqrt{3}; F_C = 10/\sqrt{3} \text{ N} = 5.77 \text{ N}$$

$$\tan D = 1; F_D = 10 \text{ N} = 10.0 \text{ N}$$

6. **Ans (C)**

$$N_1 = N_{01} e^{-\lambda_1 t}$$

$$N_2 = N_{02} e^{-\lambda_2 t}$$

$$\frac{N_1}{N_2} = \frac{2}{1} \times e^{-(\lambda_1 - \lambda_2)t}$$

$$\lambda_1 t = 3 \ln 2$$

$$\lambda_2 t = 3 \ln 2 \times \frac{3}{4.5} = 2 \ln 2$$

$$\Rightarrow \frac{N_1}{N_2} = 2 \times e^{-\ln 2} = 1$$

7. **Ans (B)**

Consider complete system's centre of mass accelerating in vertical direction with $5 \sin 37^\circ$, then

$$60 - N = 2 (5 \sin 37^\circ)$$

$$N = 54$$

8. **Ans (C)**

$$\omega = \frac{KQq}{R} + \frac{KQ(-q)}{R} + \frac{Kq^2}{a}$$

9. **Ans (B)**

$$\text{Use } \vec{U} = \vec{m} \cdot \vec{B}$$

$$KE = U_t - U_f$$

10. **Ans (D)**

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$-\frac{1}{v} + \frac{1}{u} = \frac{1}{2f}$$

$$u = \frac{400}{3} \text{ cm}$$

11. **Ans (A)**

$$\frac{f_0}{f_e} = 10, f_0 + f_e = 1.1$$

$$\text{Now } m = 10 \left[1 + \frac{f_e}{D} \right] = 14$$

12. **Ans (B)**

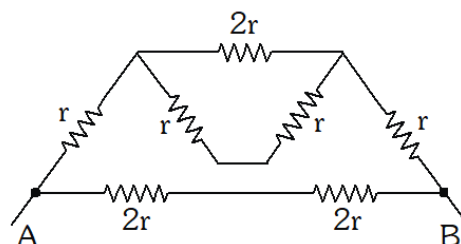
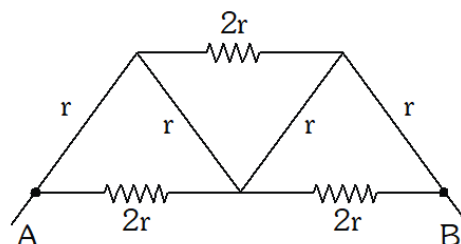
$$T = 2\pi \sqrt{\frac{\frac{ml^2}{12} + m \left(R^2 - \frac{l^2}{4} \right)}{mg \sqrt{\left(R^2 - \frac{l^2}{4} \right)}}};$$

$$l = R$$

$$= 2\pi \sqrt{\frac{\frac{mR^2}{12} + \frac{3mR^2}{4}}{mg \frac{\sqrt{3}R}{2}}} = 2\pi \sqrt{\frac{10}{12} \frac{mR^2 \times 2}{\sqrt{3}mgR}}$$

$$T = 2\pi \sqrt{\frac{5R}{3\sqrt{3}g}}$$

13. **Ans (C)**



14. **Ans (A)**

$$V_{cm} - Rw = -5$$

$$V_{cm} + Rw = 10$$

$$2Rw = 15$$

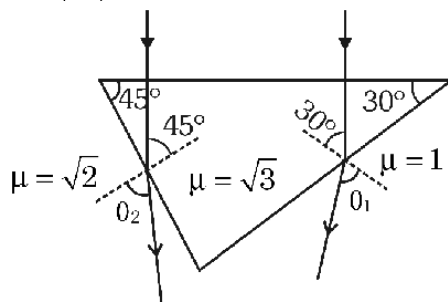
$$Rw = 7.5 \text{ cm/s}$$

$$V_{P/cm} = 10 - 2.5 = 7.5 \text{ cm/s}$$

$$V_{cm} = 2.5 \text{ cm/s}$$

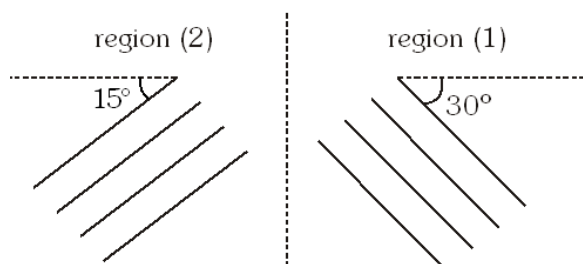
$$\text{So } s_{P/cm} = 7.5 \times 2 = 15 \text{ cm}$$

15. Ans (B)



$$\sin \theta_1 = \sqrt{3} \sin 30$$

$$\sin \theta_1 = \sqrt{3} \times \frac{1}{2}$$



$$\theta_1 = 60^\circ$$

$$\sqrt{2} \sin \theta_2 = \sqrt{3} \sin 45$$

$$\sin \theta_2 = \frac{\sqrt{3}}{2}$$

$$\theta_2 = 60^\circ$$

16. Ans (C)

$$KE_{\max} = h\nu - \frac{h\nu}{3} = \frac{2h\nu}{3}$$

$$\text{So } 0 \leq KE \leq K_{\max}$$

17. Ans (B)

The direction of $(\vec{E} \times \vec{B})$ gives the direction of propagation of wave $\hat{i} \times \hat{j} \rightarrow \hat{k}$.

18. Ans (B)

Since pressure will remain constant

$$\therefore H = C_p \frac{dT}{dt} = \frac{5R}{2} \frac{dT}{dt} \quad \dots (i)$$

From ideal gas equation

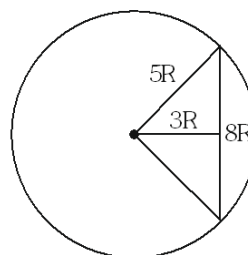
$$\left(P_0 + \frac{Mg}{A}\right) A v dt = R dT \quad \dots (ii)$$

From (i) and (ii)

$$V = \frac{2H}{5(P_0 A + Mg)}$$

19. Ans (A)

$$\phi = \frac{q_{\text{in}}}{\epsilon_0} = \frac{\rho \left(\frac{1}{2} 8R \cdot 3R\right) R}{\epsilon_0} = \frac{12\rho R^3}{\epsilon_0}$$

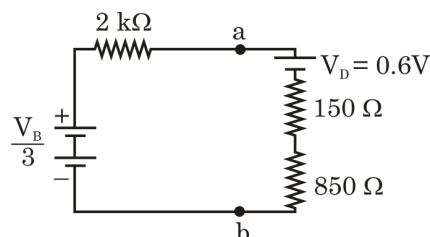


20. Ans (A)

As shown in the Figure, the diode is replaced by its equivalent circuit and the circuit to the left of the terminals a, b. Since the diode can dissipate a maximum power of 200 mW, the maximum safe diode current I will satisfy the relationship.

$$p = 200 \times 10^{-3} = i^2 r = 150 i^2$$

$$\Rightarrow i = \sqrt{\frac{0.2}{150}} = 0.0365 \text{ A} = 36.5 \text{ mA.}$$



As shown in the Figure,

$$i = \frac{(V_B/3) - 0.6}{3 \times 10^3} = 36.5 \times 10^{-3}$$

$$\Rightarrow V_B = 330 \text{ V,}$$

Which is the maximum permissible battery voltage.

PART-1 : PHYSICS

SECTION-II

1. Ans (2)

$$T = Kx$$

$$TR = \left(\frac{M_1 \ell^2}{3} + \frac{M_2 R^2}{2}\right) \alpha$$

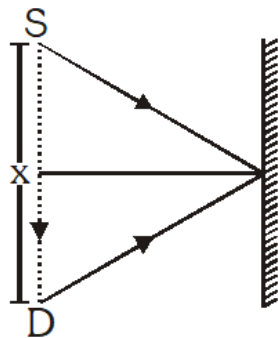
$$(KR)x = 2\alpha$$

$$\alpha = \frac{1}{2}x$$

$$\omega = \frac{1}{\sqrt{2}}$$

$$T = 2\sqrt{2}\pi$$

2. Ans (3)



The situation is shown in figure. Suppose the detector is placed at a distance of x meter from the source. The direct wave received from the source travels a distance of x meter. The wave reaching the detector after reflection from the wall has travelled a distance of $2[(2)^2 + x^2/4]^{1/2}$ meter. The path difference between the two waves is :

$$\Delta = \left\{ 2 \left[(2)^2 + \frac{x^2}{4} \right]^{1/2} - x \right\} \text{ meter}$$

Constructive interference will take place when $\Delta = \lambda, 2\lambda, \dots$. The minimum distance x for the maximum corresponds to

$$\Delta = \lambda \quad \dots(i)$$

$$\text{The wavelength is } \lambda = \frac{u}{v} = \frac{360 \text{ m/s}}{180 \text{ s}^{-1}} = 2 \text{ m}$$

$$\text{Thus, by (i), } 2 \left[(2)^2 + \frac{x^2}{4} \right]^{1/2} - x = 2$$

$$x = 3$$

Thus, the detector should be placed at a distance of 3 m from the source. Note that there is no abrupt phase change.

3. Ans (30)

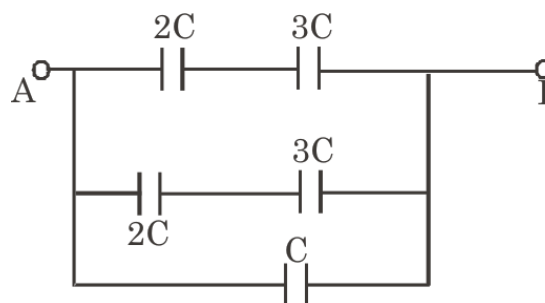
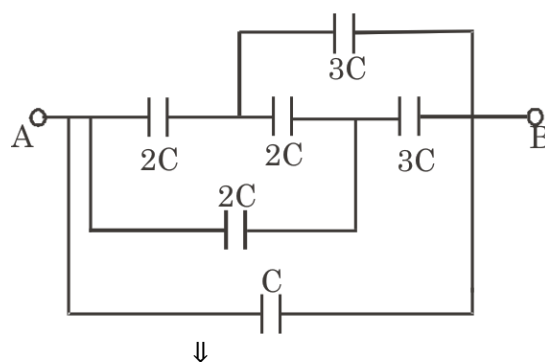
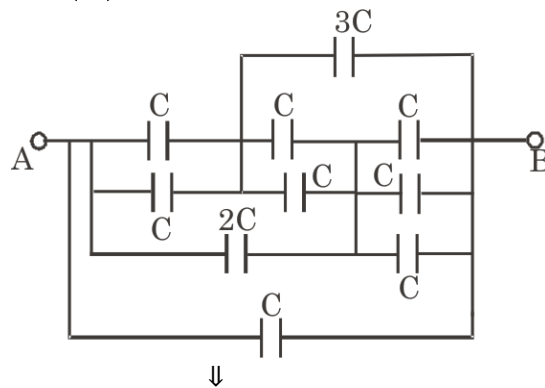
From Brewster's Law

$$\mu = \tan i_p$$

$$i_p = 60^\circ$$

$$r = 30^\circ$$

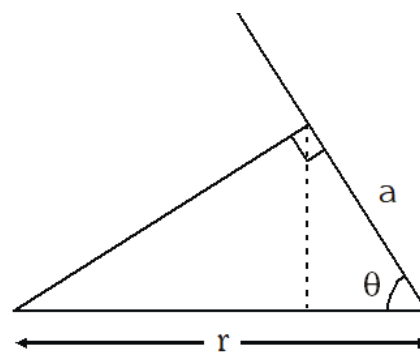
4. Ans (8)



5. Ans (60)

$$\omega r = \omega' a$$

$$v_B = \omega [r - (2a \cos \theta)] + \omega' a$$



$$\begin{aligned} &= \omega \left[r - \frac{2a^2}{r} \right] + \omega r \\ &= \omega \left[\frac{r^2 - 2a^2 + r^2}{r} \right] \\ &= 2\omega \left[\frac{r^2 - a^2}{r} \right] \end{aligned}$$

6. Ans (250)

$$a = \frac{(2m - m)g}{2m + m + \frac{1}{R^2}} = \frac{g}{5}$$

$$a_{\text{TOP}} = \sqrt{\left(\frac{g}{5}\right)^2 + \frac{\left(\frac{g}{5}t\right)^4}{R^2}}$$

7. Ans (1)

By conservation of energy,

$$E - V = ir = V^2 r$$

$$V^2 r + V - E = 0$$

$$\Rightarrow V = \frac{-1 \pm \sqrt{1 + 4rE}}{2r}$$

$$V = \frac{\sqrt{1 + 4rE} - 1}{2r} = \frac{\sqrt{1 + 4 \times 0.5 \times 1.5} - 1}{2 \times 0.5} = 1V$$

$$E - V = ir$$

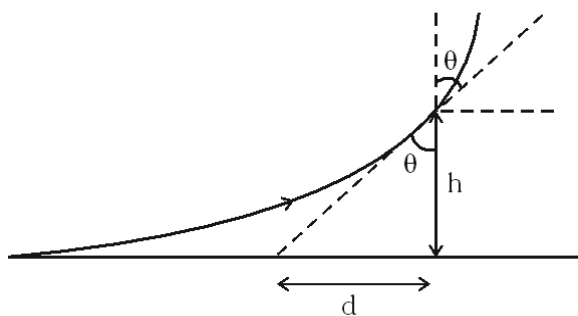
$$1.5 - 1 = i \times 0.5$$

$$i = 1A$$

8. Ans (4)

$$n = n_0(1 + ah)$$

$$n \sin \theta = n_0 \sin 90^\circ \quad (\text{by snell's law})$$



$$\Rightarrow \sin \theta = \frac{n_0}{n}$$

$$\Rightarrow \frac{d}{h} = \tan \theta$$

$$\Rightarrow d = h \tan \theta$$

$$= \frac{hn_0}{\sqrt{n^2 - n_0^2}} \quad \text{or} \quad d = \frac{h_0}{\sqrt{\left(\frac{n}{n_0}\right)^2 - 1}}$$

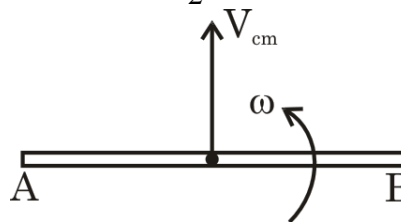
on putting values,

$$d = 750 \text{ m} = \frac{3}{4} \text{ km}$$

$$\Rightarrow p = 4$$

9. Ans (4)

$$V_A = V_{\text{cm}} - \omega \frac{\ell}{2} = 0$$



$$\Rightarrow V_{\text{cm}} = \frac{\omega \ell}{2}$$

$$\therefore T = N \times \frac{2\pi}{\omega} = \frac{2V_{\text{cm}}}{g}$$

$$H_{\text{max}} = \frac{V_{\text{cm}}^2}{2g} = \frac{Ng\pi\ell}{2 \times 2g} = \frac{N\pi\ell}{4}$$

$$K = 4$$

10. Ans (3)

When both the block in same phase only one spring exert force on block A. When out of phase two spring exert force on block A.

$$\omega_1 = \sqrt{\frac{k}{m}}, \quad \omega_2 = \sqrt{\frac{3k}{m}}$$

$$\frac{\omega_2^2}{\omega_1^2} = 3$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (D)

No reaction in option (D) as NH_2^- is poor leaving group.

3. Ans (A)

It produces CH_4 and Al_4C_3 will produce 3 moles of CH_4 per mole of Al_4C_3 .

4. Ans (A)

Cinnabar : HgS ; Malachite : $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$;
Magnesite : MgCO_3 ; Galena : PbS .

6. Ans (C)

NGP at β -position dominates.

8. Ans (D)

Size of Bk^{3+} ion is less than Np^{3+} ion is due to actinoid contraction. Actinoid contraction is the gradual decrease in the size of M^{3+} ions across the series as atomic number increases.

9. **Ans (D)**

Alkali solution dissolve impure alumina, CO_2 gas is added to complex to change pH & regenerate alumina.

10. **Ans (B)**

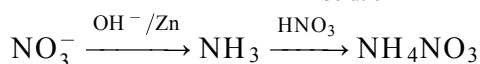
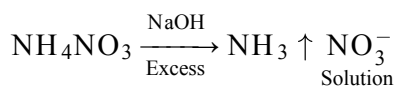
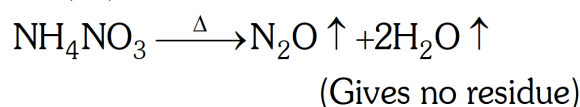
$$(\Delta_f H^0)_{\text{CO}_2(\text{g})} = -100 \text{ kcal/mole}$$

$$10 = [0.1 \times \frac{7}{2} (2 \times 10^{-3})]$$

$$\frac{10}{0.7 \times 10^{-3}} = \Delta T$$

$$\Delta T \simeq 14285.71 \text{ K}$$

11. **Ans (A)**



12. **Ans (A)**

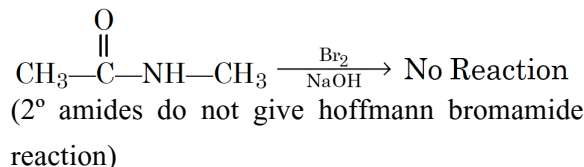
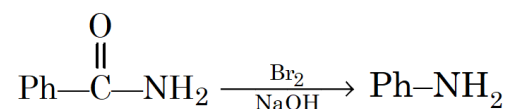
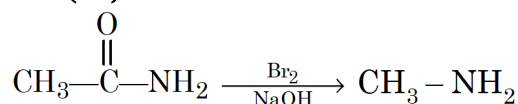
Since O^{2-} is FCC, effective number of atoms = 4

Effective number of X = 1

Effective number of Y = 2

$$\therefore \text{Fraction} = \frac{\text{Void occupied}}{\text{Void available}} = \frac{1(\text{occupied by X}) + 1(\text{occupied by Y})}{4} = \frac{1}{2}$$

13. **Ans (C)**



14. **Ans (C)**

SbH_3 has higher bp than NH_3 due to vander waal forces.

In hydracids, the acidity increases down the group while for oxyacids, the acidity decreases down the group.

15. **Ans (D)**

(A) is incorrect, because electronic transfer take place and it is chemisorbtion.

(B) is incorrect, adsorbtion is always exothermic.

(C) is incorrect, O_2 will accepted the electron from metal into its π^*2p orbital.

(D) is correct, since, the electron enters into π -orbital bond order decreases and bond length increases.

16. **Ans (C)**

CHCl_3 & Acetone shows negative deviation due to weak inter H-bonding.

17. **Ans (D)**

PH_3 (Lewis base) can react with B_2H_6 (Lewis acid).

18. **Ans (B)**

$$T_B = 350 \text{ K}$$

$$T_C = \frac{8a}{27Rb} = \frac{8}{27} \times 350\text{K}$$

So the gas can't be liquefied above 103.7 K

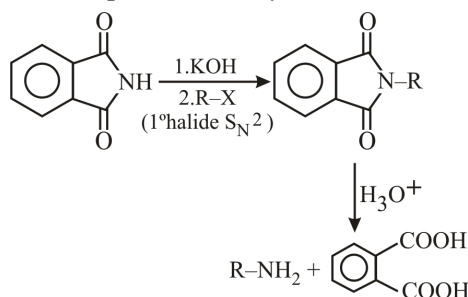
19. **Ans (D)**

It is aspirin which–

- Reduce fever (antipyretic)
- Reduce pain (non-narcotic analgesic)
- Dissolve blood clot.

20. **Ans (B)**

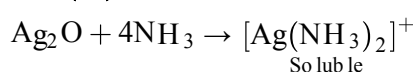
Gabriel phthalimide synthesis :



PART-2 : CHEMISTRY

SECTION-II

1. **Ans (7)**



In all other cases, precipitates are formed.

2. **Ans (100)**

Let heat involved in step 5 is Q_5

$$q_{\text{total}} = -W_{\text{total}}$$

$$500 + 800 + Q_5 = 700$$

$$Q_5 = -600 \text{ J}$$

Since $\Delta S_{\text{total}} = 0$

$$\frac{500}{250} + \frac{800}{200} - \frac{600}{T_5} = 0$$

$$2 + 4 - \frac{600}{T_5} = 0$$

$$T_5 = 100 \text{ K}$$

3. **Ans (500)**



$$t = 0 \quad 4a \quad 0 \quad 3a$$

$$t = t \quad 4a-x \quad 2x \quad 3a+x$$

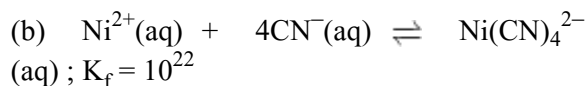
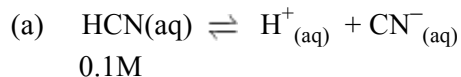
From question

$$(4a-x)(40^\circ) + 2x(10^\circ) + (3a+x)(-30^\circ) = 0^\circ$$

$$\therefore x = \frac{7}{5}a$$

$$\text{Now, } t = \frac{1}{K} \times \ln \frac{4a}{4a - \frac{7}{5}a} = \frac{1}{0.001} \times \ln \frac{20}{13} = 500 \text{ min}$$

5. **Ans (1)**



$$0.025 \text{ M} \quad 4 \times 0.025 \text{ M}$$

$$(\because V = 1\text{L}) = 0.1 \text{ M}$$

(required)

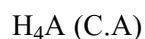
The value of K_f indicates that reaction (b) goes to completion. Thus making dissociation of HCN complete to give

$$[\text{CN}^-] = 0.1 \text{ M}$$

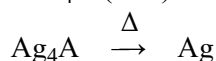
$$\therefore [\text{H}^+] = 0.1 \text{ M} = 10^{-1} \text{ M}$$

$$\therefore \text{pH} = -\log 10^{-1} = 1$$

6. **Ans (260)**



$$\therefore \text{S.S.} \Rightarrow$$



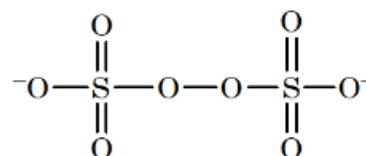
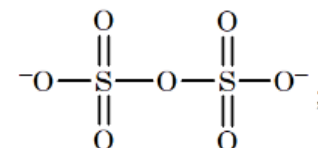
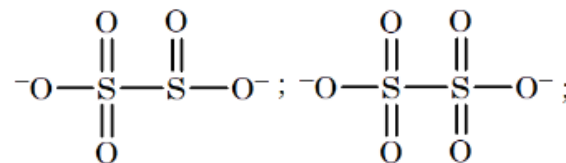
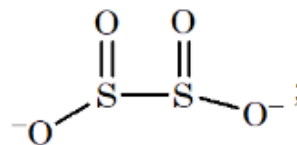
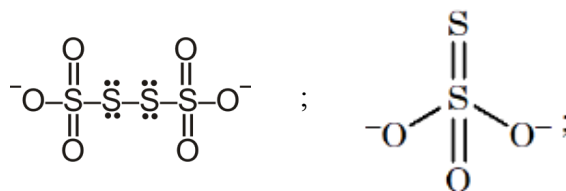
$$\text{Moles of Ag} = \frac{4.32}{108} = 4 \times 10^{-2}$$

$$\therefore \text{Moles of S.S.} = 0.01$$

$$\therefore \text{M.Mass} = 600$$

$$\therefore \text{M.M. sodium salt} = 260$$

7. **Ans (10)**



$$x = 5, y = z = 1$$

8. **Ans (300)**

MO can be reduced by C at 2000°C .

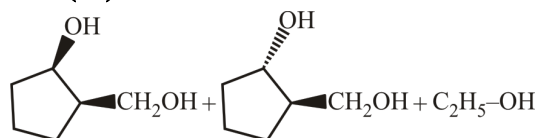


$$|\Delta G_r| = 300 \text{ kJ/mole}$$

9. **Ans (6)**

(i), (iv), (vii), (ix), (x) and (xi) are aromatic.

10. **Ans (3)**



PART-3 : MATHEMATICS

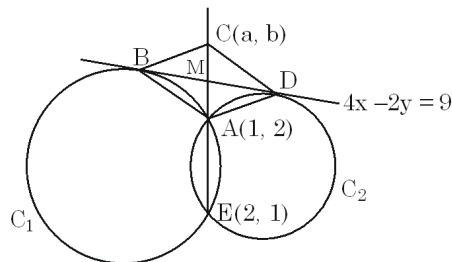
SECTION-I

1. Ans (B)

The radical axis bisects the common tangent BD.

Hence M is the mid point of BD

let C (a, b)



Now (a, b) lies on common chord AE which is

$$y - 2 = -1(x - 1)$$

$$\Rightarrow y - 2 = -x + 1$$

$$\Rightarrow x + y = 3$$

$$\therefore a + b = 3 \quad \dots(1)$$

Also M $\left(\frac{a+1}{2}, \frac{b+2}{2}\right)$ lies on $4x - 2y = 9$

$$4\left(\frac{a+1}{2}\right) - 2\left(\frac{b+2}{2}\right) = 9$$

$$\Rightarrow 2a + 2 - b - 2 = 9 \Rightarrow 2a - b = 9 \quad \dots(2)$$

Solving (1) and (2) $a = 4$ and $b = -1$

$$\Rightarrow C(a, b) \equiv C(4, -1) \text{ Ans.}$$

2. Ans (D)

$$2a + b + c = 0$$

$$\Rightarrow 4a + 2b + 2c = 0$$

$$\Rightarrow \underbrace{4a + 2b + c}_{f(2)} + \underbrace{c}_{f(0)} = 0$$

$$\text{As } f(0) + f(2) = 0$$

$$\Rightarrow \text{Either } f(0) = 0 = f(2) \text{ or } f(0) \cdot f(2) < 0$$

$$\Rightarrow \text{Either } 0 \text{ and } 2 \text{ are roots or one root lie in}$$

$$(0, 2) \text{ of } f(x) = 0.$$

$$\Rightarrow \text{Two real and distinct roots of } f(x) = 0$$

3. Ans (A)

$$u(x) = 7v(x)$$

$$\Rightarrow u'(x) = 7v'(x)$$

$$\Rightarrow p = 7 \text{ (given)}$$

$$\text{again } \frac{u(x)}{v(x)} = 7$$

$$\Rightarrow \left(\frac{u(x)}{v(x)}\right)' = 0$$

$$\Rightarrow q = 0; \text{ now } \frac{p+q}{p-q} = \frac{7+0}{7-0} = 1 \text{ Ans.}$$

4. Ans (C)

$$e_H = \frac{5}{4}; e_C = \frac{5}{3}$$

$$\text{area} = \frac{d_1 d_2}{2} = \frac{100}{2} = 50$$

$$AC: x^2 + y^2 = 16; A_H: x^2 + y^2 = 9$$

5. Ans (D)

$$\text{Given } \vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$$

$$\vec{b} \cdot \vec{c} = \vec{b} \cdot \{(2\vec{a} \times \vec{b}) - 3\vec{b}\}$$

$$\vec{b} \cdot \vec{c} = -3|\vec{b}|^2 \quad \dots(1)$$

$$\text{now } |\vec{a} \times \vec{b}|^2 = \vec{a}^2 \vec{b}^2 - (\vec{a} \cdot \vec{b})^2$$

$$= 16 - 4 = 12$$

$$\text{and } \vec{c}^2 = (2\vec{a} \times \vec{b} - 3\vec{b})^2$$

$$|\vec{c}|^2 = 4|\vec{a} \times \vec{b}|^2 + 9\vec{b}^2 - \underbrace{6\vec{b} \cdot (2\vec{a} \times \vec{b})}_{\text{zero}}$$

$$= 48 + 144 = 192$$

$$|\vec{c}| = 8\sqrt{3}$$

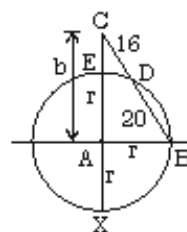
$$\text{Now, } \cos \theta = \frac{\vec{b} \cdot \vec{c}}{|\vec{b}| |\vec{c}|} = \frac{-3|\vec{b}|^2}{|\vec{b}| |\vec{c}|} = \frac{-3|\vec{b}|}{|\vec{c}|} =$$

$$\frac{-3(4)}{8\sqrt{3}} = \frac{-\sqrt{3}}{2}$$

$$\text{Hence } \vec{b} \wedge \vec{c} = \frac{5\pi}{6} \text{ Ans.}$$

6. Ans (B)

$$b^2 + r^2 = (36)^2 \quad \dots(1)$$



Also $CD \cdot CB = CE \cdot CX$ (using power of the point C)

$$16 \cdot 36 = (b-r)(b+r)$$

$$\begin{aligned} \therefore b^2 - r^2 &= 16 \cdot 36 \\ \text{from (1) and (2)} \\ 2b^2 &= 36(36 + 16) = 36 \cdot 52 \\ b^2 &= 36 \cdot 26 \\ \Rightarrow b &= 6\sqrt{26} \text{ Ans.} \end{aligned}$$

7. **Ans (A)**

$$\begin{aligned} x^{2 \ln x - 1} + e^{1/9} &= (1 + e^{1/9}) \left(x^{\frac{2 \ln x - 1}{2}} \right) \\ \text{Let } x^{\frac{2 \ln x - 1}{2}} &= y \\ \therefore y^2 + e^{1/9} &= (1 + e^{1/9})y \\ y^2 - (1 + e^{1/9})y + e^{1/9} &= 0 \\ 2y &= (1 + e^{1/9}) \pm \sqrt{1 + e^{2/9} + 2e^{1/9} - 4e^{1/9}} \\ 2y &= (1 + e^{1/9}) \pm (1 - e^{1/9}) \end{aligned}$$

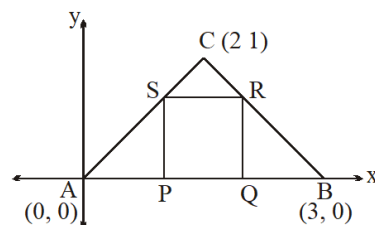
8. **Ans (C)**

$$\begin{aligned} \text{Let } \theta &= \frac{\pi}{4} + x \\ \Rightarrow d\theta &= dx \\ \text{or } 4\theta &= \pi + 4x \\ \Rightarrow \pi - 4\theta &= -4x \\ I &= \int_{-\pi/2}^0 \frac{(-4x) \tan(\frac{\pi}{4} + x)}{1 - \tan(\frac{\pi}{4} + x)} dx = -4 \int_{-\pi/2}^0 \frac{x(1 + \tan x)}{1 - \tan x} dx = -4 \\ &\int_{-\pi/2}^0 \frac{x(1 + \tan x)}{1 - \tan x} \cdot \frac{(1 - \tan x)}{(-2) \tan x} dx \\ &= 2 \int_{-\pi/2}^0 \frac{x(1 + \tan x)}{\tan x} dx = 2 \int_{-\pi/2}^0 \left(\frac{x}{\tan x} + x \right) dx \\ I &= x^2 \Big|_{-\pi/2}^0 + \int_{-\pi/2}^0 \frac{x}{\tan x} dx = -\frac{\pi^2}{4} + 2 \int_0^{\pi/2} \frac{t}{\tan t} dt \\ x &= -t \\ \text{now } I_1 &= \int_0^{\pi/2} \underbrace{t}_{I} \underbrace{\cot t}_{II} dt = [t \ln \sin t]_0^{\pi/2} - \\ &\int_0^{\pi/2} \ln \sin t dt \quad I_1 = 0 + \frac{\pi}{2} \ln 2 \\ \text{Hence } I &= 2 \cdot \frac{\pi}{2} \ln 2 - \frac{\pi^2}{4} = \pi \ln 2 - \frac{\pi^2}{4} \end{aligned}$$

9. **Ans (B)**

Let PQRS be the square inscribed in $\triangle ABC$ with length of each side equal to a . Let the coordinates of P be $(p, 0)$. So, the coordinates of Q, R, S are respectively $(p + a, 0)$, $(p + a, a)$ and (p, a) .

Now, equation of AC is $y = \frac{1}{2}x$ and S (p, a) lies on it.



$$\Rightarrow a = \frac{p}{2} \text{ or } p = 2a$$

Also equation of BC is $(x + y) = 3$ and

R $(p + a, a)$ lies on it.

$$\begin{aligned} \therefore (p + a) + a &= 3 \Rightarrow 4a = 3 \Rightarrow a = \frac{3}{4} \text{ and } \\ p &= \frac{3}{2} \\ \text{So, } P\left(\frac{3}{2}, 0\right), Q\left(\frac{9}{4}, 0\right), R\left(\frac{9}{4}, \frac{3}{4}\right) \text{ and } S\left(\frac{3}{2}, \frac{3}{4}\right). \end{aligned}$$

Centre of square PQRS

$$= \left(\frac{\frac{3}{2} + \frac{9}{4}}{2}, \frac{0 + \frac{3}{4}}{2} \right) = \left(\frac{15}{8}, \frac{3}{8} \right)$$

Ans.

10. Ans (C)

We have

$\tan^{-1} x + \cot^{-1} x = \sin^{-1} x + \cos^{-1} x$, which is true for all $x \in [-1, 1]$

So, number of integral values of $x = 3$.

$$\vec{c} = |\vec{a} \times \vec{b}| \vec{a} + (\vec{a} \cdot \vec{b}) \vec{b}, |\vec{a}| = 1 = |\vec{b}|$$

$$\vec{a} \wedge \vec{b} = \frac{\pi}{6}$$

$$\therefore |\vec{a} \times \vec{b}| = (1)(1) \sin \frac{\pi}{6} = \frac{1}{2} \vec{a} \cdot \vec{b} = (1)(1)$$

$$\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \vec{c} = \frac{1}{2} \vec{a} + \frac{\sqrt{3}}{2} \vec{b}$$

$$\Rightarrow |2\vec{c}|^2 = |\vec{a} + \sqrt{3}\vec{b}|^2$$

$$\Rightarrow 4|\vec{c}|^2 = 1 + 3 + 2\sqrt{3} \left(\frac{\sqrt{3}}{2} \right) = 7$$

$$\therefore 4|\vec{c}|^2 = 7$$

$$\text{Let } I = \int_{-\pi/2}^{\pi/2} \frac{\ln(\sec x)}{(2013)^{\sin x} + 1} dx$$

$$\text{So, } I = \int_0^{\pi/2} \ln(\sec x)$$

$$\left[\frac{1}{(2013)^{\sin x} + 1} + \frac{1}{(2013)^{-\sin x} + 1} \right] dx$$

$$= - \int_0^{\pi/2} \ln(\cos x) dx = \frac{\pi}{2} \ln 2$$

$$\therefore k = 2$$

11. Ans (D)

$$\text{Given, } x^2 - 6x + 8 \leq 0 \Rightarrow (x-2)(x-4) \leq 0$$

$$\Rightarrow x \in [2, 4]$$



$$\text{Also, } x^2 - 2mx + m^2 - 4 \leq 0 \Rightarrow (x-m)^2 - 4 \leq 0$$

$$-2 \leq x - m \leq 2$$

$$m - 2 \leq x \leq m + 2 \Rightarrow x \in [m-2, m+2]$$

$$m + 2 \geq 2 \text{ \& } m - 2 \leq 4$$

$$m \geq 0 \text{ \& } m \leq 6$$

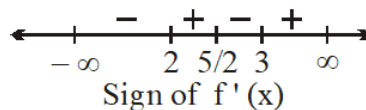
$$\therefore m \in [0, 6]$$

12. Ans (A)

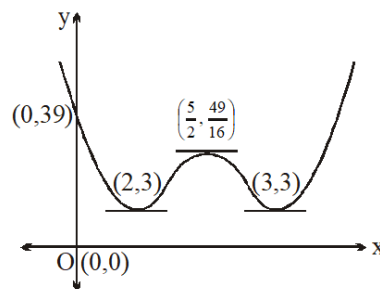
$$\text{Let } f(x) = k(x-2)^2(x-3)^2 + 3$$

$$\text{As, } f(1) = 4k + 3 = 7$$

$$\therefore k = 1 \Rightarrow f(x) = (x-2)^2(x-3)^2 + 3$$



$$\text{Now, } f'(x) = 2(x-2)(x-3)(2x-5)$$



$$\Rightarrow f(x) \text{ has local maximum at } x = \frac{5}{2}$$

$$\therefore f_{\max.} \left(x = \frac{5}{2} \right) = \frac{49}{16}$$

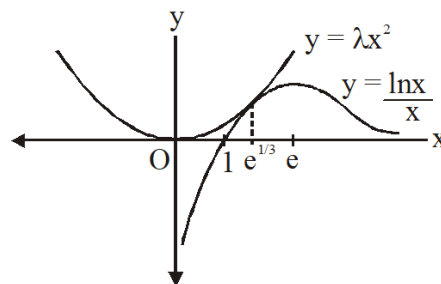
13. Ans (B)

Since, C_1 and C_2 touches each other.

$$\Rightarrow \frac{\ln x}{x} = \lambda x^2$$

$$\Rightarrow \ln x = \lambda x^3 \quad \dots(1)$$

$$\text{and } \frac{1 - \ln x}{x^2} = 2\lambda x$$



$$\Rightarrow 1 - \lambda x^3 = 2\lambda x^3$$

$$\Rightarrow 3\lambda x^3 = 1$$

$$\Rightarrow x = \left(\frac{1}{3\lambda} \right)^{1/3} = (3\lambda)^{-1/3}$$

$$\text{From (1), } \frac{-1}{3} \ln(3\lambda) = \lambda \cdot \frac{1}{3\lambda}$$

$$\Rightarrow \ln(3\lambda) = -1. \Rightarrow \lambda = \frac{1}{3e} \quad \text{Ans.}$$

14. Ans (B)

$$\underbrace{X_1 X_2 \dots X_{19} X_{20}}_{2 \text{ from } 19} \underbrace{X_{21} X_{22} \dots X_{50}}_{3 \text{ from } 30}$$

$$\text{Total ways } n(E) = {}^{19}C_2 \times {}^{30}C_3$$

$$\therefore \text{ Required probability} =$$

$$\frac{n(E)}{n(S)} = \frac{{}^{19}C_2 \times {}^{30}C_3}{{}^{50}C_5}$$

15. Ans (A)

$$\lim_{n \rightarrow \infty} \left(\frac{n!}{(mn)^n} \right)^{1/n}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1.2.3.4 \dots (n-1)n}{n^n} \right)^{1/n} \times \frac{1}{m}$$

$$P = \lim_{n \rightarrow \infty} \left\{ \left(\frac{1}{n} \right) \left(\frac{2}{n} \right) \left(\frac{3}{n} \right) \dots \left(\frac{n-1}{n} \right) \left(\frac{n}{n} \right) \right\}^{1/n}$$

$$\log_e P = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \log_e \left(\frac{r}{n} \right) + \log_e \frac{1}{m}$$

$$= \int_0^1 \log_e x \, dx - \log_e m$$

$$= -1 - \log_e m$$

$$\log_e P = -\log_e e - \log_e m$$

$$= -\log_e (em)$$

$$= \log_e \left(\frac{1}{em} \right)$$

$$\therefore P = \frac{1}{em}$$

16. Ans (B)

$$f\{g(s)\} = \begin{cases} 1, & 0 < x < \frac{3\pi}{4} \text{ or } \frac{7\pi}{4} < x < 2\pi \\ 0, & x = \frac{3\pi}{4}, \frac{7\pi}{4} \\ -1, & \frac{3\pi}{4} < x < \frac{7\pi}{4} \end{cases}$$

Clearly $f\{g(x)\}$ is not continuous at

$$x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

17. Ans (C)

$$\text{Let } p(x) = ax^4 + bx^3 + cx^2 + dx + e$$

$$p(2) = 16a + 8b + 4c + 2d + e = -1$$

$$p'(x) = 4ax^3 + 3bx^2 + 2cx + d$$

$$p'(2) = 32a + 12b + 4c + d = 0$$

$$p''(x) = 12ax^2 + 6bx + 2c$$

$$p''(2) = 48a + 12b + 2c = 2$$

$$p'''(x) = 24ax + 6b$$

$$\therefore p'''(2) = 48a + 6b = -12$$

and

$$p^{iv}(x) = 24a \Rightarrow p^{iv}(2) = 24a \Rightarrow 24 = 24a$$

$$\therefore a = 1$$

$$\text{From Eq. (iv), } b = -10$$

$$\text{From Eq. (iii), } c = 37$$

$$\therefore p''(x) = 12x^2 - 60x + 74$$

$$\therefore p''(1) = 12 - 60 + 74$$

$$= -48 + 74 = 26$$

18. Ans (A)

$$\text{Let point on parabola } y^2 = 8x \text{ be } (2\alpha^2, 4\alpha)$$

$$\text{Centre of circle} = (0, -6)$$

Distance between centre of circle and point on parabola

$$S = \sqrt{(2\alpha^2)^2 + (4\alpha + 6)^2}$$

$$D = 4\alpha^2 + (4\alpha + 6)^2$$

$$\text{Where } D = S^2$$

$$\frac{dD}{d\alpha} = 16\alpha^3 + 2(4\alpha + 6) \cdot 4 = 0$$

$$\Rightarrow 2\alpha^3 + 2\alpha + 3 = 0$$

$$\Rightarrow (\alpha^2 - \alpha + 3)(\alpha + 1) = 0$$

$$\Rightarrow \alpha = -1 \quad (\because \alpha^2 - \alpha + 3 \neq 0)$$

$$\text{Now, } \frac{d^2D}{d\alpha^2} = 48\alpha^2 + 8.4$$

$$= 48\alpha^2 + 32 > 0 \text{ at } \alpha = 1$$

$\Rightarrow D$ is minimum at $\alpha = -1$. Point on parabola $(2, -4)$

19. Ans (A)

$$\begin{aligned}\text{Let } I &= \int \frac{(x^4 - x)^{1/4}}{x^5} dx \\ &= \int \frac{x \left(1 - \frac{1}{x^3}\right)^{1/4}}{x^5} dx \\ &= \int \frac{\left(1 - \frac{1}{x^3}\right)^{1/4}}{x^4} dx \\ \text{Put } 1 - \frac{1}{x^3} &= t^4 \Rightarrow \frac{3}{x^4} dx = 4t^3 dt \\ \therefore I &= \frac{4}{3} \int_0^x t \cdot t^3 dt = \frac{4}{3} \cdot \left(\frac{5}{5}\right) + c \\ &= \frac{4}{15} \left(1 - \frac{1}{x^3}\right)^{5/4} + c\end{aligned}$$

20. Ans (C)

$$\begin{aligned}\frac{1}{y(\ln y)^2} \frac{dy}{dx} + \frac{1}{\ln y} \times \frac{1}{x} &= \frac{1}{x^2} \\ \text{Put } \frac{1}{\ln y} &= t, \text{ So, } \frac{-1}{y(\ln y)^2} \frac{dy}{dx} = \frac{dt}{dx} \\ \text{Hence, DE becomes} \\ \frac{dt}{dx} - \frac{t}{x} &= \frac{-1}{x^2} \Rightarrow \text{IF} = \frac{1}{x} \\ \text{So, } \frac{t}{x} &= - \int \frac{1}{x^3} dx + C = \frac{1}{2x^2} + C \\ \Rightarrow \frac{1}{\ln y} &= \frac{1}{2x} + Cx\end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (3600)

Consonant $\rightarrow$ b, c, d, f, g, h (6)

Vowels $\rightarrow$ a, e, i, o, u (5)

w :

×		×		×		×
---	--	---	--	---	--	---

Case I: If word begins with consonants

$$\text{then } ({}^6C_4 \times 4!) \times ({}^5C_3 \times 3!) = 360 \times 60 = 21600$$

Case II : If word begins with vowels

$$({}^5C_4 \times 4!) \times ({}^6C_3 \cdot 3!) = 120 \times 120 = 14400$$

$$\text{Total} = 36000 \Rightarrow 10K = 36000 \Rightarrow K = 3600$$

2. Ans (212)

Equation of the plane P is

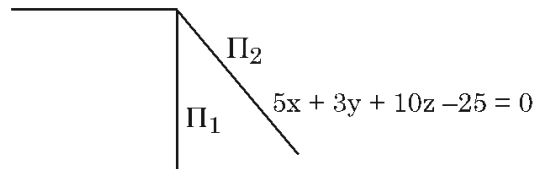
$$4x + 7y + 4z + 81 + \lambda(5x + 3y + 10z - 25) = 0$$

$$(5\lambda + 4)x + (3\lambda + 7)y + (10\lambda + 4)z +$$

$$(81 - 25\lambda) = 0$$

$$\text{Now } 4(5\lambda + 4) + 7(3\lambda + 7) + (10\lambda + 4)4 = 0$$

Π



$$4x + 7y + 4z + 81 = 0$$

$$81\lambda = -81 \Rightarrow \lambda = -1$$

equation of the plane

$$-x + 4y - 6z + 106 = 0$$

$$p = \left| \frac{106}{\sqrt{1+16+36}} \right| = \left| \frac{(53)(2)}{\sqrt{53}} \right| = \sqrt{212}$$

$$\Rightarrow k = 212$$

3. Ans (6)

$$\text{We have } T_r - 3T_{r-1} = 6^r$$

Dividing both sides by 3^r , we get

$$\frac{T_r}{3^r} - \frac{T_{r-1}}{3^{r-1}} = 2^r$$

$$\Rightarrow \sum_{r=2}^n \left(\frac{T_r}{3^r} - \frac{T_{r-1}}{3^{r-1}} \right) = \underbrace{\sum_{r=2}^n 2^r}_{\text{G.P.}}$$

$$\left. \begin{aligned} &\frac{T_2}{3^2} - \frac{T_1}{3} \\ &+ \frac{T_3}{3^3} - \frac{T_2}{3^2} \\ &+ \frac{T_4}{3^4} - \frac{T_3}{3^3} \\ &\vdots \\ &+ \frac{T_n}{3^n} - \frac{T_{n-1}}{3^{n-1}} \end{aligned} \right\} = \sum_{r=2}^n 2^r$$

$$\begin{aligned} \frac{T_n}{3^n} - \frac{T_1}{3} &= 4(2^{n-1} - 1) \\ \frac{T_n}{3^n} - \frac{6}{3} &= 4(2^{n-1} - 1) \quad (\because T_1 = 6) \\ \Rightarrow T_n &= 2 \cdot 3^n + 4 \cdot 3^n \cdot 2^{n-1} - 4 \cdot 3^n = 2^{n+1} \cdot 3^n - 2 \cdot 3^n \\ \Rightarrow T_n &= 2(6^n - 3^n) \\ \text{So, } S_n &= 2 \left(\sum_{1}^n 6^n - \sum_{1}^n 3^n \right) = \\ &= 2 \left[\frac{6}{5} (6^n - 1) - \frac{3}{2} (3^n - 1) \right] \\ \Rightarrow S_n &= \frac{3}{5} [4 \cdot 6^n - 5 \cdot 3^n + 1] \\ \text{Now on comparing, we get} \\ \frac{(n^2 - 12n + 39)(4 \cdot 6^n - 5 \cdot 3^n + 1)}{5} &= \\ \frac{3}{5} [4 \cdot 6^n - 5 \cdot 3^n + 1] \\ \Rightarrow n^2 - 12n + 39 &= 3 \Rightarrow (n-6)^2 = 0 \\ \text{Hence } n &= 6 \end{aligned}$$

4. **Ans (3)**

We have, $f(x) = e^{2x} - (c+1)e^x + 2x + \cos 2 + \sin 1$

$$\therefore f'(x) = 2e^{2x} - (c+1)e^x + 2$$

Now, $f(x) \geq 0 \forall x \in \mathbb{R}$

$$\text{i.e., } 2 \left(e^x + \frac{1}{e^x} \right) - (c+1) \geq 0 \forall x \in \mathbb{R}$$

$$\Rightarrow (c+1) \leq \underbrace{2 \left(e^x + \frac{1}{e^x} \right)}_{\text{minimum value} = 4} \forall x \in \mathbb{R}$$

$$\therefore c+1 \leq 4 \Rightarrow c \leq 3$$

$$\therefore c \in (-\infty, 3] \equiv (-\infty, \lambda]$$

Hence $\lambda = 3$. Ans.

5. **Ans (10)**

Equation of line passing through $(5, 1, a)$ and $(3, b, 1)$ is $\frac{x-5}{2} = \frac{y-1}{1-b} = \frac{z-a}{a-1} = \lambda$ (say)

If the line cross yz -plane,

$$\text{i.e. } x=0 \Rightarrow x=2\lambda+5=0 \Rightarrow \lambda = \frac{-5}{2}$$

$$\text{Since, } y = \lambda(1-b) + 1 = \frac{17}{2}$$

$$(\text{Given}) \Rightarrow \frac{-5}{2}(1-b) + 1 = \frac{17}{2} \Rightarrow b=4$$

$$\text{Also, } z = \lambda(a-1) + a = \frac{-13}{2}$$

$$(\text{Given}) \Rightarrow \frac{-5}{2}(a-1) + a = \frac{-13}{2} \Rightarrow a=6$$

$$\text{Hence } (a+b) = 6+4 = 10$$

6. **Ans (7)**

Given curves are

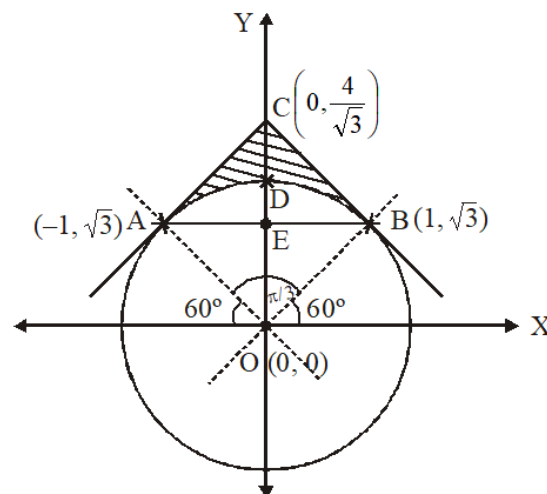
$$x^2 + y^2 = 4 \text{ and } y = \sqrt{3}|x|$$

$$\Rightarrow |x|^2 + 3|x| - 4 = 0$$

$$\Rightarrow x = \pm 1 \text{ and } y = \sqrt{3}$$

Now, equations of tangents are

$$x + \sqrt{3}y = 4 \text{ and } -x + \sqrt{3}y = 4;$$



The tangents intersects on y -axis at

$$\left(0, \frac{4}{\sqrt{3}}\right)$$

Now,

$$\Rightarrow \angle AOB = \frac{\pi}{3}$$

$$\text{So, required area} = \left(\frac{4}{\sqrt{3}} - \frac{2\pi}{3} \right) \equiv$$

$$\left(\frac{a}{\sqrt{b}} - \frac{2\pi}{3} \right)$$

$$\therefore a=4, b=3$$

$$\Rightarrow (a+b)_{\text{Least}} = 7$$

7. Ans (4)

$$\therefore \frac{x^2}{a^2} + \frac{y^2}{4} = 1 \quad \dots (i)$$

$$\text{and } y^3 = 16x \quad \dots (ii)$$

From Eq. (i),

$$\frac{2x}{a^2} + \frac{2y}{4} \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{4x}{a^2y} \quad \dots (iii)$$

From Eq. (ii),

$$3y^2 \frac{dy}{dx} = 16$$

$$\therefore \frac{dy}{dx} = \frac{16}{3y^2} \quad \dots (iv)$$

$$\left(-\frac{4x}{a^2y}\right) \left(\frac{16}{3y^2}\right) = -1$$

$$\text{or } 64x = 3a^2y^3 \\ = 3a^2(16x) \quad (\text{from Eq. (ii)})$$

$$\therefore a^2 = \frac{4}{3}$$

8. Ans (45)

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta}$$

$$= \frac{\frac{y}{z} + \frac{x}{z}}{1 - \frac{y}{z} \cdot \frac{x}{z}} = \frac{\frac{3}{6} + \frac{2}{6}}{1 - \frac{3}{6} \cdot \frac{2}{6}} = 1$$

$$\Rightarrow a + b = \angle QPR = \frac{\pi}{4}$$

9. Ans (6)

The slope of the chord is $m = -\frac{8}{y}$

$$\Rightarrow y = \pm 1, \pm 2, \pm 4, \pm 8$$

But (8, y) must also lie inside the circle

$$x^2 + y^2 = 125$$

$$\Rightarrow y \text{ can be equal to } \pm 1, \pm 2, \pm 4 \Rightarrow 6 \text{ values}$$

10. Ans (15)

$$|\vec{c} - \vec{a}| = \sqrt{14}$$

$$\Rightarrow |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c} \cdot \vec{a} = 14 \quad \dots (1)$$

$$\vec{a} \cdot \vec{c} + 2|\vec{c}| = 0$$

$$\Rightarrow |\vec{a}| \cdot |\vec{c}| \cdot \cos \theta + 2|\vec{c}| = 0$$

$$\Rightarrow |\vec{c}| \cdot (|\vec{a}| \cdot \cos \theta + 2) = 0$$

$$\Rightarrow \cos \theta = -\frac{2}{3}, \text{ given } |\vec{a}| = 3.$$

from (i)

$$\Rightarrow |\vec{c}|^2 + 9 - 2|\vec{c}| \cdot |\vec{a}| \cdot \left(-\frac{2}{3}\right) - 14 = 0$$

$$\Rightarrow |\vec{c}|^2 + 4|\vec{c}| - 5 = 0 \Rightarrow |\vec{c}| = 1, -5$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$|(\vec{a} \times \vec{b}) \times \vec{c}| = |(\vec{a} \times \vec{b})| \cdot |\vec{c}| \cdot \sin \theta$$

$$= 3.1 \times \frac{1}{2} = \frac{3}{2}$$